

INNOVATIVE DESIGN PROJECT

1. STEP 1 - STUDENT AUTHENTICATION AND PROFILES

1.2 Student enters name, age, grade.

1.2 Backend stores data in database and provides Student ID

TECHSTACK -

- React (Form)
- Express (REST API)
- Firestore

2. STEP 2 - AUDIO RECORDING

2.1 Start → Request Microphone permission → Browser returns
MediaStream

→ MediaRecorder records audio → Stop

TECHSTACK -

- React
- navigator.mediaDevices.getUserMedia()
- MediaRecorder

3. STEP 3 - BROWSER API s INVOLVED?

3.1 MediaDevices API - Requests microphone permission

3.2 MediaRecorder - Starts recording and collects audio (.webm)

3.3 Web Audio API - Analyses sound, estimates loudness, counts pause,
speaking duration

3.4 Fetch API - Uploads audio files to express server

4. STEP 4 - BACKEND PROCESSING

4.1 Object Creation-From the audio (.webm) file (from Web Audio API)
object is created that holds values → duration, pausecount, silence

4.2 JSON -The object is sent (no audio file) via the Fetch API to the Express
server

4.3 Rule Engine - Score point evaluation given to each parameter

For eg, if duration < 4 → +20

Hence, overall score is calculated

4.4 Firestore - The overall score is permanently stored in this cloud database

5. STEP 5 - DISPLAY

To display all backend processing on the frontend interface in a formatted manner

SYSTEM ARCHITECHTURE

